

AI-aided structure-based drug discovery

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Structure-based virtual screening (SBVS) is widely used to identify candidate ligands through receptor-based computational screening. However, conventional docking approaches are constrained by limited receptor flexibility, simplified scoring functions, weak agreement with experimental binding affinities, and poor scalability to ultra-large chemical libraries. Recent advances in deep learning and structure prediction, including AlphaFold2, provide new opportunities to overcome these limitations. This presentation introduces an AI-enhanced SBVS framework integrating multi-state ensemble screening, graph-based binding affinity prediction, and million-scale docking. The proposed approaches achieved performance comparable to or exceeding state-of-the-art methods, supporting more accurate and efficient hit identification.